Coulomb-metric point selection in interpolative separable density fitting

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Interpolative separable density fitting (ISDF) factorizes the two-electron Coulomb tensor at a cost that scales with the number of interpolation points $N_\mu = c_\mu N_{\rm orb}$, with every downstream object in the auxiliary index costing $O(N_\mu^2)$ in storage and $O(N_\mu^3)$ in dense linear algebra. Standard ISDF selects interpolation points by a pivoted Cholesky factorization of the orbital-pair overlap metric, which optimally resolves pair densities in the L^2 sense but over-resolves short-wavelength structure that the Coulomb kernel suppresses. We show that, because the grid metric cancels exactly from the ISDF least-squares solve, the metric can influence the factorization *only* through the choice of interpolation points and through pair-space weights, and we introduce three composable, input-driven modifications of the selection: (i) separable pair-space weighting of the selection Gram matrix, (ii) a Gaussian-filtered orbital surrogate, and (iii) an exact two-pass Coulomb-metric re-ranking of an inflated candidate pool. None of them alter the ISDF ansatz, the interpolation-vector solve, or any downstream consumer, and each reduces to the standard algorithm in a well-defined limit. On silicon we measure the error in the Hartree, exact-exchange, and RPA correlation energies as a function of c_μ and find [TODO: summary of findings: reduction of c_μ^* at fixed accuracy, per knob and composed]. The re-ranking pass adds only a factor $s \sim 2$ to the selection cost, which is itself a small fraction of the total calculation.

I. INTRODUCTION

The interpolative separable density fitting (ISDF), or tensor hypercontraction (THC), representation of the two-electron repulsion integrals underlies low-scaling implementations of Hartree–Fock exchange, MP2 and RPA correlation energies, coupled-cluster methods, *GW* and related many-body perturbation frameworks, and quantum-algorithm resource estimates [TODO: cite: Lu-Ying 2015, Hu-Lin-Yang 2017, Parrish-Hohenstein-Martinez THC, Malone-Lee QC resource, CoQui]. In all of these, the auxiliary (interpolation-point) index carries the factorization,

$$\rho_p(\mathbf{r}) \equiv u_i^*(\mathbf{r}) u_a(\mathbf{r}) \;\approx\; \sum_{\mu=1}^{N_\mu} \zeta_\mu(\mathbf{r})\, \rho_p(\mathbf{r}_\mu), \tag{1}$$

so that any object stored on the auxiliary index costs $O(N_\mu^2)$ and any factorization or inversion there costs $O(N_\mu^3)$. The proportionality constant $c_\mu = N_\mu/N_{\rm orb}$ at fixed target accuracy is therefore the single most important number in the method: reducing c_μ from 10 to 6 is a factor 2.8 in auxiliary storage and 4.6 in dense linear algebra, everywhere at once.

Standard ISDF chooses the points $\{\mathbf{r}_\mu\}$ by pivoted Cholesky factorization of the pair-density overlap (Gram) matrix, which minimizes the L^2 error of the fitted pair densities greedily. That objective treats all Fourier components of the pair density equally, while every consumer of the factorization contracts it against the Coulomb kernel $v(\mathbf{G}) = 4\pi/|\mathbf{G}|^2$ (or a screened variant), which discounts short-wavelength errors by two powers of $|\mathbf{G}|$. The overlap-selected point set consequently spends resolution where the target quantity does not need it. [TODO: 1-2 sentences on prior work on weighted/centroid selection: centroidal Voronoi ISDF, machine-learned selection, if citing.]

In this paper we adapt the point selection — and only the point selection — to the Coulomb metric. The starting observation (Sec. II A) is that the ISDF interpolation vectors are independent of any grid-space metric: the metric cancels identically from the least-squares solve. All freedom therefore resides in (a) *which* points are selected and (b) pair-space weights. We exploit this with three independent, composable knobs (Sec. **??**), all reusing the existing pivoted-Cholesky machinery, all controlled from the input file, and all reducing bitwise to the standard algorithm in their default settings. Section III measures the resulting error in the Hartree, exact-exchange, and RPA correlation energies of silicon as a function of c_μ, for norm-conserving, ultrasoft, and PAW pseudopotentials.

II. THEORY

A. ISDF and the exact cancellation of the grid metric

Collect the pair densities on the real-space grid in $Z \in \mathbb{C}^{N_g \times N_{\rm pair}}$, $Z_{gp} = \rho_p(\mathbf{r}_g)$, and let $\Theta = Z[I,:]$ be its restriction to the interpolation points $I = \{\mathbf{r}_\mu\}$. The ISDF ansatz (1) is $Z \approx \zeta\,\Theta$ with $\zeta \in \mathbb{C}^{N_g \times N_\mu}$. Minimizing the weighted residual in an arbitrary (Hermitian, positive definite) grid metric v and pair weight $W = {\rm diag}(w_p)$,

$$E \;=\; {\rm Tr}\big[(Z - \zeta\Theta)\, W\, (Z - \zeta\Theta)^\dagger\, v\big]\,, \tag{2}$$

gives the stationarity condition $v\,(Z - \zeta\Theta)\, W\, \Theta^\dagger = 0$, and since v is invertible,

$$\zeta \;=\; Z\, W\, \Theta^\dagger \left(\Theta\, W\, \Theta^\dagger\right)^{-1}. \tag{3}$$

The grid metric v *cancels identically*; the pair weight W does not. The structural reason is that ζ carries a free grid index, so the residual is driven orthogonal to the row space of Θ at every grid point independently. Three consequences shape everything that follows: the ζ solve already in production is v-optimal for the given points; no alternating or quartic-scaling optimization can improve it at fixed I; and the metric can act only through the selection of I and through W.

B. The selection objective

With ζ eliminated, the fit error in the metric v as a function of the point set is

$$E_v(I) \;=\; \mathrm{Tr}\,[\,Q_\perp\, M\,], \qquad M = Z^\dagger v\, Z, \tag{4}$$

with $Q_\perp$ the projector onto the orthogonal complement of the selected rows in pair space. Greedy minimization of (4) is pivoted Cholesky on

$$K \;=\; Z\,M\,Z^\dagger \;=\; C\,v\,C, \qquad C = ZZ^\dagger, \tag{5}$$

where $C_{gg'} = A_{gg'} \odot B^*_{gg'}$ has the Hadamard structure of a product of orbital Gram matrices, $A_{gg'} = \sum_i u_i(\mathbf{r}_g)u_i^*(\mathbf{r}_{g'})$, that makes its diagonal and columns computable in $O(N_g N_{\mathrm{orb}})$. Standard selection replaces $M \to \nVdash$, i.e. pivots on C itself. Pivoting on K directly is not affordable: v is a convolution and does not commute with the Hadamard structure, so a single column of K costs $O(N_g N_i N_a)$ and the full selection $O(N_g N^3_{\mathrm{orb}})$. The three knobs below approach the v-weighted objective at essentially the cost of the standard one.

C. Knob 1: separable pair-space weights

[TODO: condense spec Sec. 2: separable weights preserve the Hadamard structure at $N_L\times$ cost; PSD by the Schur product theorem for $f_p, g_p, c_p \geq 0$; Laplace expansion for energy denominators; selection-only by default; rotation-invariance caveat.]

D. Knob 2: filtered-orbital surrogate

For point selection only, attenuate each orbital's plane-wave coefficients,

$$\tilde{c}_{n\mathbf{k}}(\mathbf{G}) = c_{n\mathbf{k}}(\mathbf{G})\, e^{-\alpha|\mathbf{k}+\mathbf{G}|^2/G_c^2}, \tag{6}$$

with G_c the wavefunction cutoff, build the selection Gram from the $\tilde{u}$, pivot, then solve Eq. (3) with the *unfiltered* orbitals. Filtering both factors of the pair density attenuates its transform at roughly $\sqrt{2}$ smaller width with a Gaussian rather than $1/G^2$ profile, so this is a surrogate for — not an implementation of — the Coulomb metric; its virtue is that it costs one extra FFT sweep and nothing else. $\alpha = 0$ recovers the standard algorithm exactly.

FIG. 1. [TODO: Error in E_x, E_H, E_{RPA} vs c_μ for Si (NCPP), baseline vs knobs.]

E. Knob 3: exact Coulomb re-ranking of a candidate pool

Run the standard (or knob-1/2–modified) pivoted Cholesky to an inflated rank $N_1 = \lceil sN_\mu \rceil$, $s \in [1.5, 3]$, producing a candidate pool P and the Cholesky factor $L \in \mathbb{C}^{N_g \times N_1}$. Because the Schur complement vanishes exactly on eliminated pivots, $C[P,:] = L[P,:]\,L^\dagger \equiv \ell L^\dagger$ with $\ell = L[P,:]$ lower triangular, one has, exactly,

$$K_{PP} \;=\; \ell\,(L^\dagger v L)\,\ell^\dagger \;=\; Y^\dagger Y, \qquad Y = R\,\ell^\dagger, \tag{7}$$

where $R(\mathbf{G}, j) = \sqrt{\bar{v}(\mathbf{G})}\,\hat{L}(\mathbf{G}, j)$ is assembled from the Fourier transforms of the Cholesky columns and a pluggable multiplier $\bar{v}(\mathbf{G})$ (bare, attenuated, $\mathbf{q}$-averaged, or screened). A dense pivoted Cholesky of the $N_1 \times N_1$ matrix K_{PP} then re-ranks the pool in the Coulomb metric and keeps the leading N_μ points. The v-optimum is searched only within the L^2-good pool — re-ranking, not re-searching — which is what makes the cost a factor $\sim s$ on selection with no change in scaling. $s = 1$ and $\bar{v} = 1$ are exact no-op limits.

F. Implementation in CoQui

[TODO: brief: pivoted-Cholesky driver anchors; options table (the nine isdf_* keys with defaults); bitwise-regression defaults; PAW consistency — augmentation channels already Coulomb-ranked, point selection now matches; cost table from spec Sec. 4.4.]

III. RESULTS

[TODO: Computational details: Si fcc, 2x2x2 (and 4x4x4 production) MP mesh, NCPP/USPP/PAW, cutoffs, beta=1000 DLR, reference = Cholesky 1e-10 (NCPP) and THC self-convergence (USPP/PAW). Error metrics: —dE$_H|, |dE_x|, |dE_R PA| vs c_m u; c_m u * tables; per - knob scans (alpha, s, gcut, N_L) and composition.$]

IV. DISCUSSION AND CONCLUSIONS

[TODO: when does the metric matter (vacuum, localized orbitals, metals); go/no-go criterion from spec Sec. 8; screened-interaction extension; composition guidance and recommended defaults.]

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